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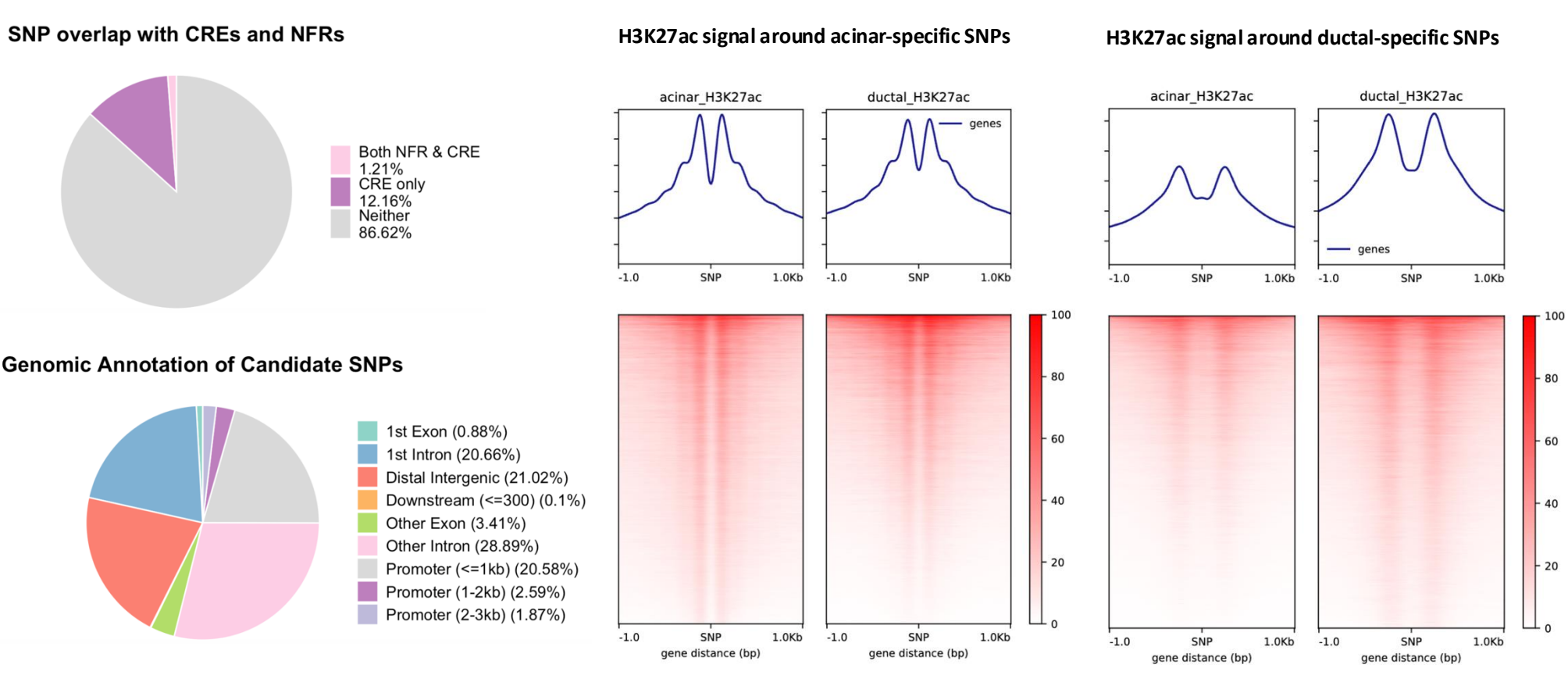
Background

Pancreatitis is a complex inflammatory disorder of the pancreas with both environmental and genetic contributors. Genome-wide association studies (GWAS) have identified multiple single nucleotide polymorphisms (SNPs) associated with pancreatitis susceptibility, yet the functional relevance of many of these variants remains unclear. Notably, a large proportion of GWAS-identified risk SNPs are located in non-coding regions, suggesting potential roles in gene regulation through disruption of cis-regulatory elements (CREs). This study aims to systematically identify functional SNPs associated with pancreatitis risk and elucidate their regulatory mechanisms using integrated multi-omics approaches.

Methods

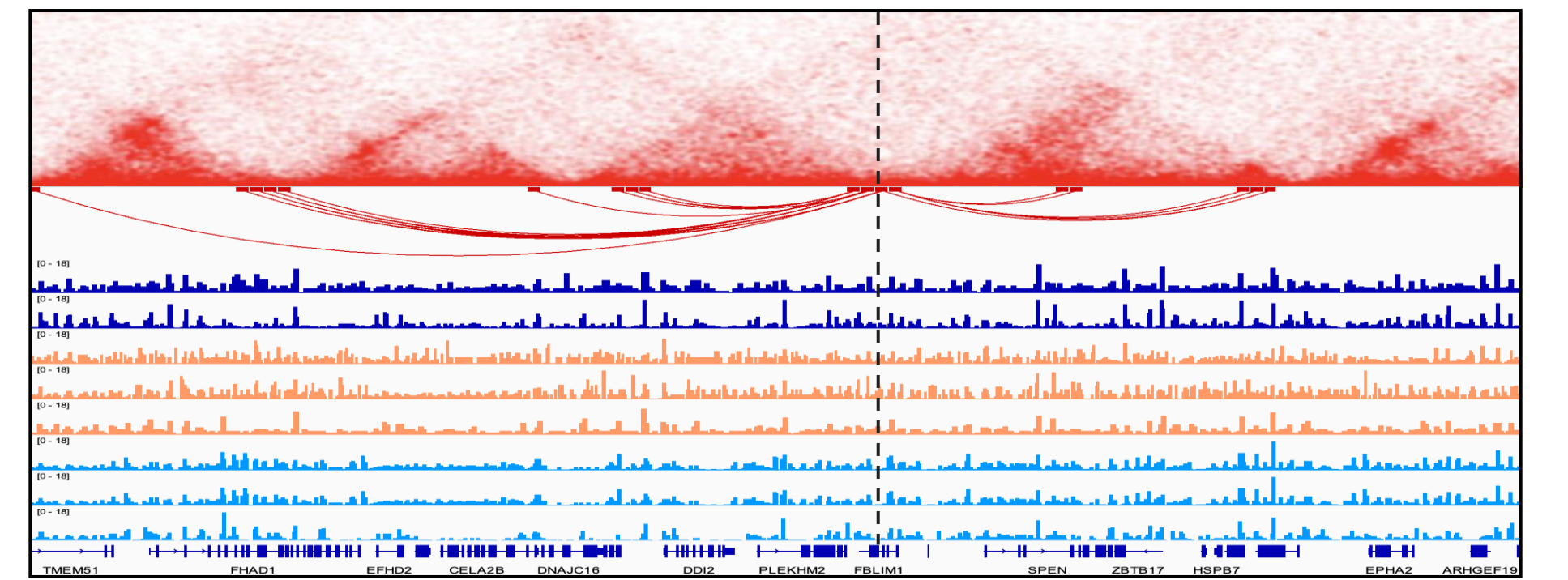
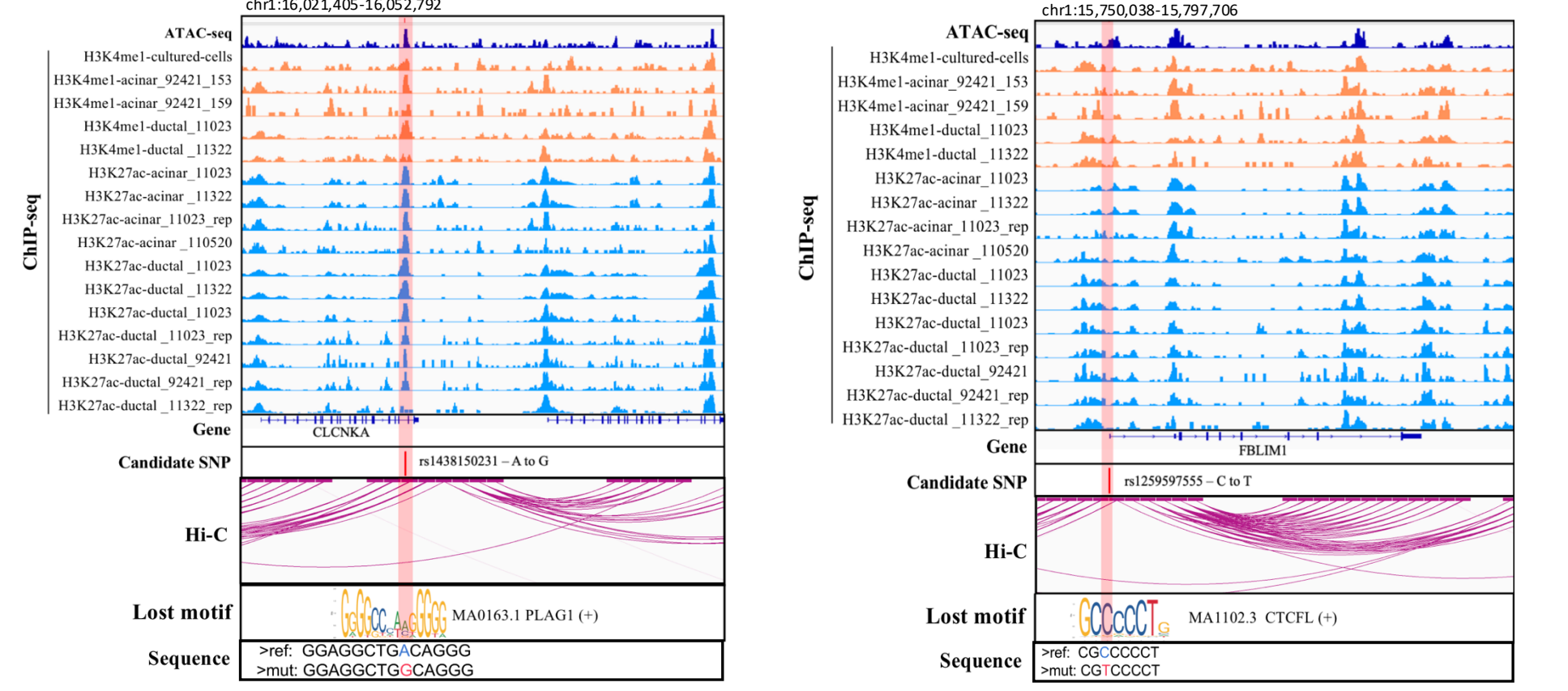
Risk SNPs associated with pancreatitis were screened using the GWAS catalog. For each SNP, we analyzed all variants within a ± 10 Mb window to capture proximal candidate variants. To define CREs, we used ATAC-seq and ChIP-seq data from human pancreatic acinar and ductal cells. Candidate functional SNPs were selected based on overlap with cis-regulatory elements and nucleosome-free regions (NFRs)-overlapping proximal variants from the ± 10 Mb scanned regions. Chromatin interaction data from Hi-C were used to link SNP-associated regulatory elements to their potential target genes. Selected regulatory elements were perturbed using CRISPR interference (CRISPRi), followed by transcriptome profiling. Phenotypic effects of prioritized SNPs were further evaluated in a mouse model of pancreatitis.

Functional annotation of pancreatitis-associated SNPs

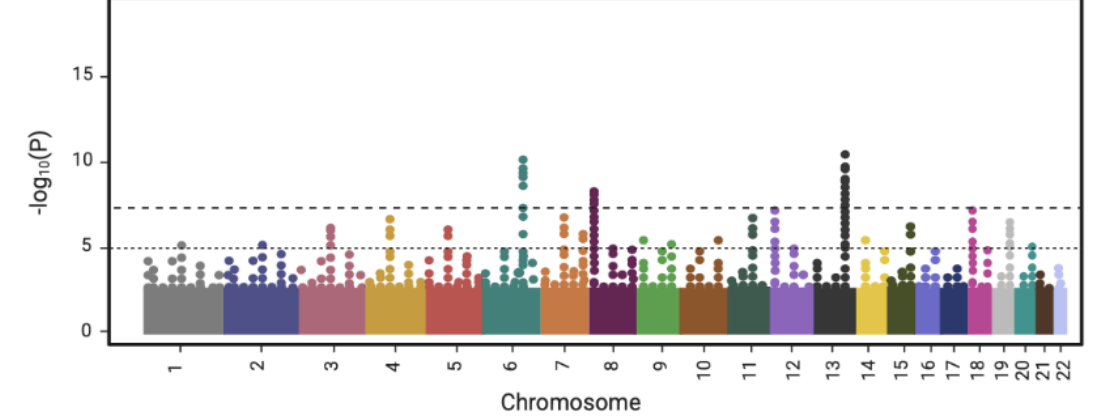


We collected 204 pancreatitis risk SNPs from 12 GWAS studies and scanned ± 10 Mb around each, identifying ~ 205 million surrounding SNPs. Among them, 27.41 million (13.38%) overlapped with CREs, and only 2.48 million (1.21%) overlapped both NFRs and cCREs. In total, we identified 2,914,754 SNPs as our candidate variants. We further annotated the genomic context of candidate SNPs, revealing substantial enrichment in non-coding regions, consistent with potential regulatory function. Heatmaps and average profiles represent H3K27ac enrichment ± 1 kb around SNPs specifically associated with acinar or ductal cell NFRs. Each row in the heatmap corresponds to a SNP-centered region.

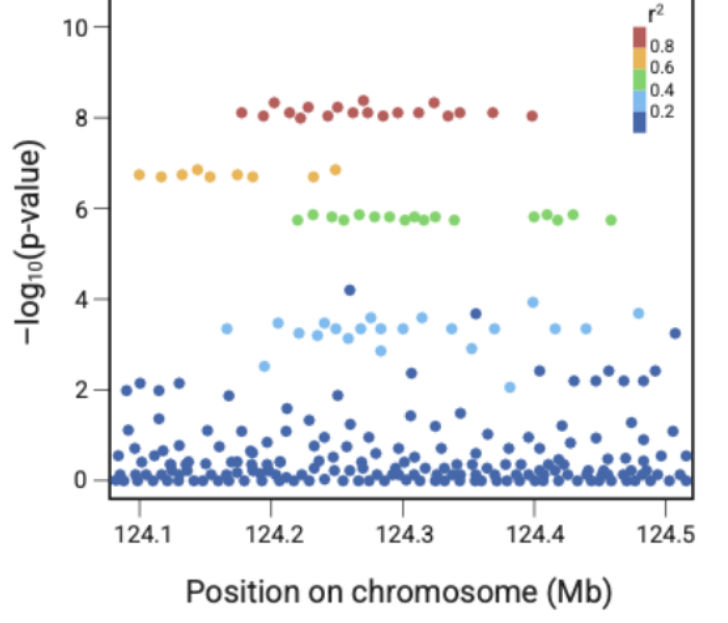
rs125959755 disrupts CTCF binding sites, alters TAD boundary and long-range Enhancer-Promoter loops



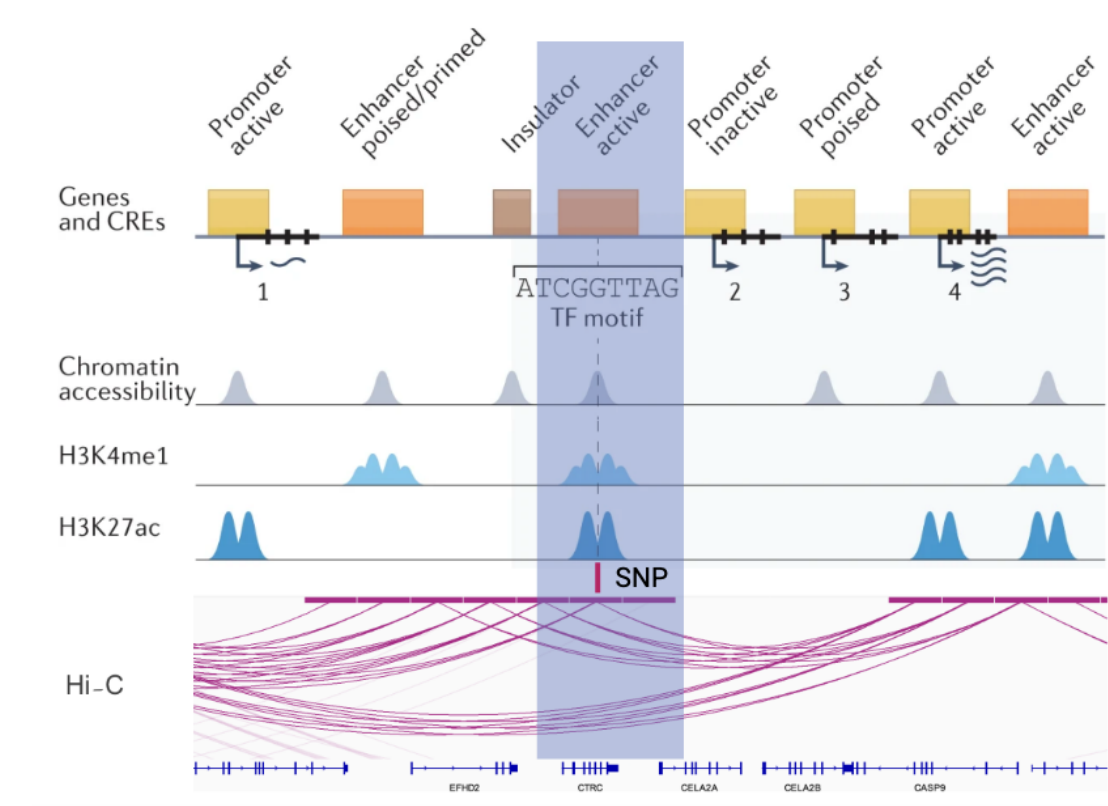
Screening for pancreatitis-associated SNPs from previous GWAS studies



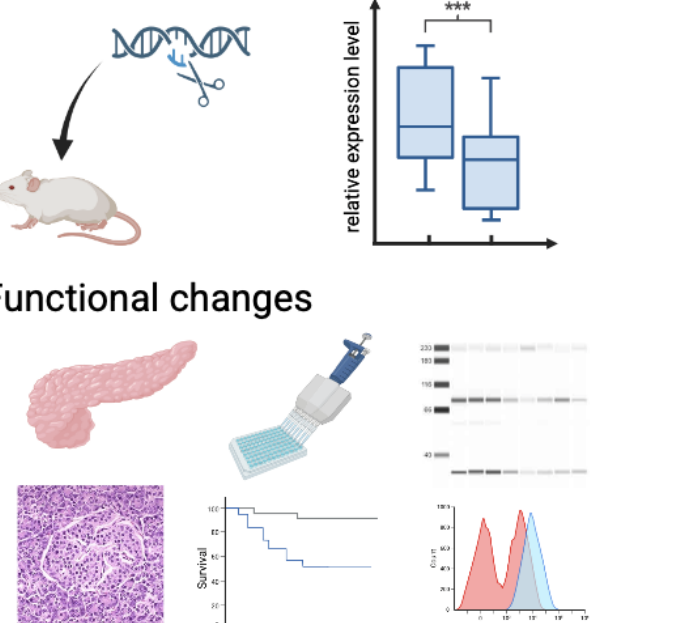
LD analysis to identify potentially causative SNPs



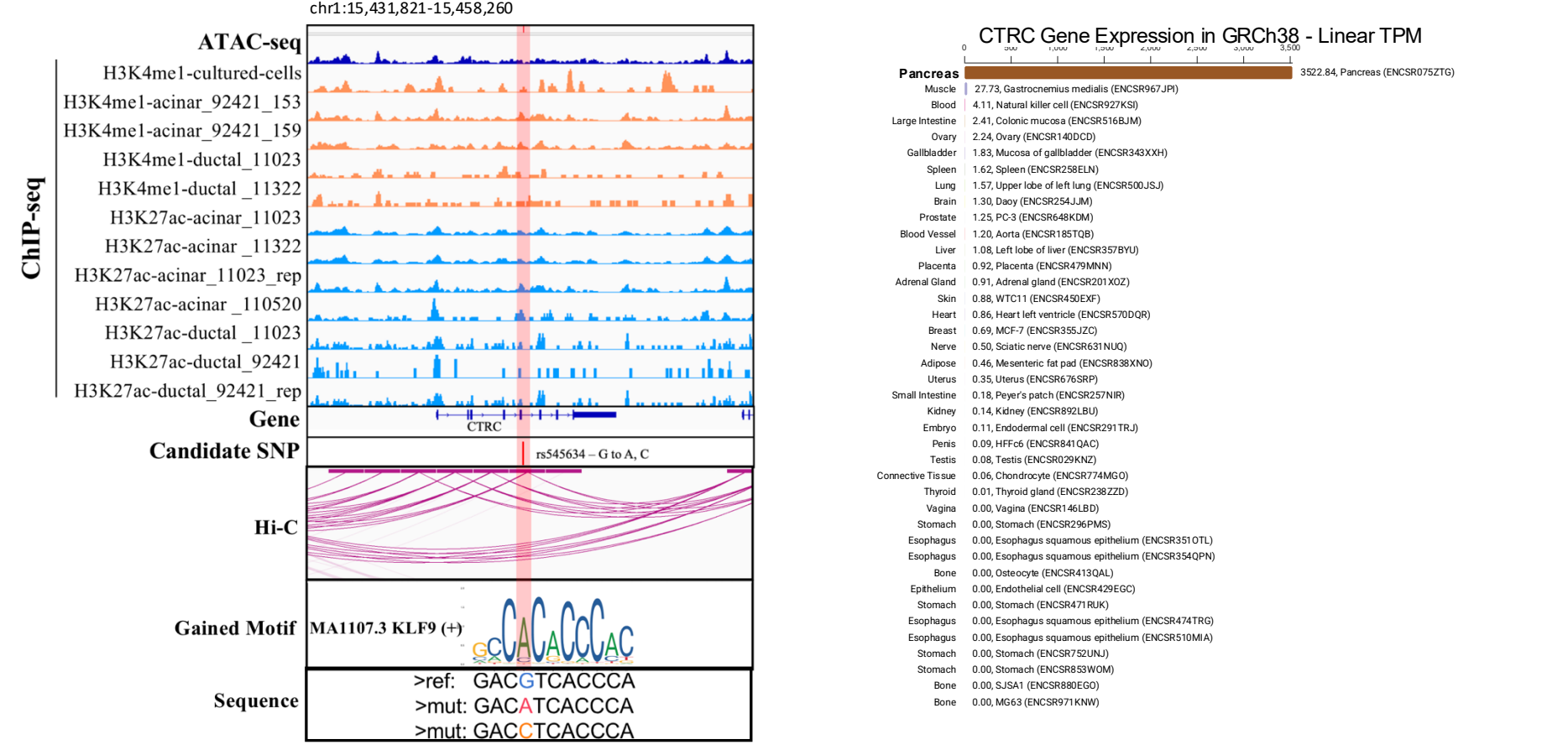
ATAC-seq, ChIP-seq for cCREs identification



Functional validation with CRISPRi



Gained KLF9 binding motif may alter pancreas-specific expression gene, CTRC



Using FIMO to perform motif analysis on candidate SNPs, we identified a gained motif for KLF9 (MA1107.3) at the site of rs545634 (G>A/C). The gain motif is located at the CTRC gene position, which is highly specifically expressed in the pancreas (TPM > 3500). These findings suggest that rs545634 may influence pancreas-specific gene regulation by creating a novel transcription factor binding site near CTRC.

Acknowledgement
NIH/NCI CA218968, CA237159, NIH/NIDDK R01DK110361, CPRIT First-Time, Tenure-Track Faculty Award (RR240016), UT Rising STARS (UT175673), MCC-American Cancer Society Early Career pilot award.

Conclusion

Our integrative analysis identifies candidate functional SNPs that may contribute to pancreatitis risk by altering transcription factor binding and 3D chromatin structure in pancreatic cells.

Future Directions

Given the conserved 3D genome structure at the CTCFL motif locus between human and mouse, we plan to establish a mouse model to validate the regulatory function of the candidate variant in vivo.